MTH220 2022 S1 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ICT133 Structured Programming
* ICT162 Object Oriented Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

Donaldson Tan

MTH220

Timed Online Assignment – January Semester 2022

Statistical Methods and Inference

Thursday, 26 May 2022 **1:00 pm – 3:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **SIX (6)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O)*.

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown .

Answer all questions. (Total 100 marks)

Question 1

(a) In a study to estimate the percentage of citizens who feel financial security is an important aspect of having money, 938 adults were polled. Describe the following:

(i) the quantity of interest;

(ii) the population;

(iii) the sample;

(iv) the population parameter; and

(v) the sample statistic.

(5 marks)

(b) The financial literacy scores of citizens are normally distributed with mean 75 and standard deviation 8. A selected group of 20 retirees have an average score of 70.

(i) Compute the probability that 20 citizens selected at random will have a mean score of less than 70. Show full details of your workings.

(ii) Comment on the level of financial literacy of this selected group of retirees.

(iii) Comment whether the Central Limit Theorem is necessary in the above calculation. Support your answer with brief reasons.

(10 marks)

Question 2

A laboratory technician wants to compare the breaking strength of three kinds of thread. He obtained the breaking strengths (g) as follows:

Thread type 1: $x_{11}, x_{12}, x_{13}, x_{14}, x_{15}, x_{16}$
Thread type 2: $x_{21}, x_{22}, x_{23}, x_{24}, x_{25}, x_{26}$
Thread type 3: x_{31}, x_{32}, x_{33}

(a) Compute the missing entries and help the technician complete the ANOVA table in Table Q2(a).

Source of variation	**Degrees of freedom**	**Sum of Squares**	**Mean Squares**	**F-value**
Treatments				
Error			1.86	
Total		37.46		

Table Q2(a)

(8 marks)

(b) Displaying all necessary details, perform an analysis of variance to test at 0.05 level of significance, whether the differences among the sample means are significant.

(7 marks)

Question 3

Over a period, it has been found that the average number of cavities that a child has per year is 0.36. Cavoff Company has developed a new toothpaste. A random sample of 40 children who used the new Cavoff toothpaste for a year was taken. This sample had a mean of 0.31 cavities and a standard deviation of 0.15 cavities. Apply an appropriate hypothesis test at 5% level, on the claim that children who use Cavoff toothpaste have better yearly dental check-up results. Show full details of the test carried out.

(15 marks)

Question 4

In a study to compare the average number of years working in the company by salespersons and office personnel, two random samples were selected. The first sample of 12 salespersons has a mean of 3.2 and standard deviation of 0.6, while the second sample of 19 office personnel has a mean of 5.6 and standard deviation of 0.2.

(a) Perform an appropriate hypothesis test to determine if office personnel tend to stay longer, at 5% level of significance. Show in details all the relevant steps in the test.

(15 marks)

(b) Determine the expression of the p-value for the test, which need not be evaluated. Hence comment if its value will yield similar conclusion as in Question 4(a).

(5 marks)

Question 5

It is thought that basketball teams that makes too many fouls in a game tend to lose the game even if they play well. A study was carried out with the following results:

x, number of fouls more than opposing team	0	2	5	6
y, percentage of times, the team with large number of fouls wins the game	50	45	33	26

Table Q5

(a) Plot a scatter diagram and comment on the relationship between x and y.

(5 marks)

(b) Given that $SS_x \approx 22.75,\ \overline{x} = 3.25$, apply appropriate formulation to verify the following.

(i) $SS_{xy} \approx -89.5$

(ii) The estimated least-square equation line is given by $y = 51.286 - 3.934x$.

(7 marks)

(c) If a team committed 4 fouls more than the opposing team, what is the estimated probability of the team losing the game?

(3 marks)

Question 6

Table Q6 below shows the Myers-Briggs personality performance and profession types of a random sample of 2408 representatives.

	Personality Profession Type	
Profession Type	**Extroverted**	**Introverted**
Clergy	308	226
Medical Doctor	667	936
Lawyer	112	159

Table Q6

Carry out an appropriate test to determine that the listed profession types and personality preferences are independent at 0.01 level of significance.

Writing out **details** (together with the details of any needed calculation) of

(a) the hypotheses,

(2 marks)

(b) derivations of the expected frequencies,

(6 marks)

(c) the test statistic and its distribution

(7 marks)

(d) critical value and

(2 marks)

(e) conclusion.

(3 marks)

----- END OF PAPER -----

1(a) The quantity of interest is characteristic of a population we would like to study

The population refers to the set of possible cases we would like to make a statement

Sample is a subset of the population of which data is collected from

Population parameter is a numerical quantity that describes a population.
E.g. population variance.

Statistic is a numerical quantity that can be measured from sample data

(b) let X be the financial literacy score of a citizen
$X \sim N(75, \sigma^2)$

let $\bar{X}$ be the mean score of a sample of size 20

$\bar{X}_n \sim N(75, \frac{\sigma^2}{20})$

i. $P(\bar{X} < 70) = \text{pnorm}(70, 75, \frac{\sigma}{\sqrt{20}})$
$= 0.00259$

ii. The selected group has a financial literacy that is very below average.
Their financial literacy ranks at 0.259 percentile of the population.

iii. The population is already normally distributed, so central limit theorem is not required.

2(a)

Source of variations	Degree of Freedom	Sum of Squares	Mean Squares	F-value
Treatments	2	37.46 − 22.32 = 15.14	15.14 ÷ 2 = 7.57	7.57 ÷ 1.86 = 4.07
Error	12	12 × 1.86 = 22.32	1.86	
Total	14	37.46		

$N = 6+6+3 = 15$ $\quad N-K = 12$

$K = 3$ $\quad K-1 = 2$

(b): $H_0: \mu_1 = \mu_2 = \mu_3$

$H_1: \mu_1 \neq \mu_2 \neq \mu_3$

At $\alpha = 0.05$ with $df_{numerator} = 2$, $df_{denominator} = 12$

$F_{critical} = qf(0.95, 2, 12)$

$= 3.885294$

$F > F_{critical} \Rightarrow$ reject null hypothesis.

$\therefore$ The difference among sample means is significant

3. $H_0: \mu = 0.36$

$H_1: \mu < 0.36$

$\Rightarrow$ use lower-tail t-test.

at $\alpha = 0.05$, $t_{critical} = qt(\alpha, n-1)$ where $n = 40$

$= qt(0.05, 39)$

$= -1.684875$

$\bar{x} = 0.31$, $s = 0.15$, $n = 40$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{0.31 - 0.36}{\frac{0.15}{\sqrt{40}}} = -2.108185$$

since $t < t_{critical}$, there is sufficient evidence that children who use Cavoff toothpaste have better yearly dental checkup.

4. (a) let group 1 represents salesperson
let group 2 represents office personnel

$H_0: \mu_2 = \mu_1$

$H_1: \mu_2 > \mu_1$

$n_1 = 12, \ \bar{x}_1 = 3.2, \ s_1 = 0.6$

$n_2 = 19, \ \bar{x}_2 = 5.6, \ s_2 = 0.2$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

$$= 0.1616743$$

$$s_p = 0.4017204$$

$$t = \frac{(\bar{x}_2 - \bar{x}_1) - (\mu_2 - \mu_1)}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = 16.2022$$

At $\alpha = 0.05$, $t_{critical} = qt(1-\alpha, n_1+n_2-2)$

$= qt(0.95, 29)$

$= 1.699127$

since $t > t_{critical}$, there is sufficient evidence that office personnel tends to stay longer

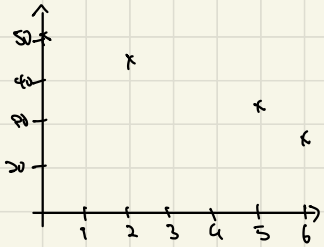
(b) p-value $= 1 - P(T \leq t)$

$= 1 - P(T \leq 16.2022)$

since $t > t_{critical}$, then the p-value will be less than α

Hence, the null hypothesis is rejected, reaching the same conclusion as (a).

5 (a)



x and y appears to share a linear relationship whereby y decreases while x increases.

(b) $\Sigma x = 13 \quad \Sigma y = 13 \quad \Sigma xy = 411 \quad n = 4$

$$SS_{xy} = \Sigma xy - \frac{1}{n}\Sigma x \Sigma y$$
$$= -89.5$$

$$B_1 = \frac{SS_{xy}}{SS_{xx}} = \frac{-89.5}{22.75} = -3.934$$

$$B_0 = \bar{y} - B_1\bar{x}$$
$$= \frac{1}{n}\Sigma y - \frac{B_1}{n}\Sigma x$$
$$= 51.286$$

$$\therefore \quad y = B_0 + B_1 x$$
$$= 51.286 - 3.934x$$

(c) $\hat{y}(x) = B_0 + B_1 x$

$$\hat{y}(4) = B_0 + B_1(4)$$
$$= 35.55$$

estimated probability $= 0.3555$

6(a) H_0: Personality & Profession Type are independent

H_1: Personality & Profession Type are not independent.

total number of observations $N = 2408$

1) Table for Observed Frequencies

		1 Extroverted	2 Introverted	Total per Profession
1	clergy	308	226	534
2	medical doctor	667	936	1603
3	lawyer	112	159	271
	total by personality	1087	1321	

where $cell(i,j) = O(i,j)$

2) Table for Expected Frequencies:

		1 Extroverted	2 Introverted
1	clergy	241.0548	292.9452
2	medical doctor	723.6134	879.3866
3	lawyer	122.3326	148.6674

where $cell(i,j) = E(i,j) = \frac{r_i \cdot c_j}{N}$

r_i is the total frequence for row i in table 1

c_j is the total frequence for col j in table 1

$N = 2408$

3) Table for Partial Chi-Square

		1 Extroverted	2 Introverted
1	clergy	18.5923837	15.2989578
2	medical doctor	4.429263 0	3.6446668
3	lawyer	0.8729369	0.7181366

where $cell(i,j) = \frac{(O(i,j) - E(i,j))^2}{E(i,j)}$

$$\chi^2 = \sum_{i=1}^{3} \sum_{j=1}^{2} \frac{(O(i,j) - E(i,j))^2}{E(i,j)} = 43.55614$$

$df = (3-1)(2-1) = 2$

at $\alpha = 0.01$, $\chi^2_{critical} = qchisq(1-\alpha, df) = qchisq(0.99, 2) = 9.21034$

since $\chi^2 > \chi^2_{critical}$, there is sufficient evidence to conclude that personality type and profession type are not independent.